

OS
Assignment - 01

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21/08/23

• Sec-A (5*2)

Q 1. What is Operating System?

Ans. Operating system is an interface between the hardware and user. An operating system is responsible for management and coordination of activities, sharing of resources of the computers.
ex: Windows, Linux, Ubuntu, Android, etc.

Q2. Mention the components of OS.

Ans. Components of Operating system are:-

- (i) Kernel (ii) Shell (iii) File system (iv) Process management, (v) Memory management (vi) Device Driver (vii) User interface (viii) Network Management

Q3. What is difference between Program and Process?

Ans. A Program is a set of instructions written to perform a specific task. It is a passive entity; that is stored in secondary memory; such as hard drive or SSD.

→ A process is an instance of a program that is being executed. It is an active entity that is allocated resources by the OS, such as CPU time, memory and I/O devices.

Program	Process
• Program is a passive entity. • It resides in the secondary memory of the system. • It does not have a resource requirement. • It doesn't have a control block.	• Process is a active entity. • It is created when a program is in execution and is being loaded into the main memory. • It has a high resource requirement. • It has its own control block, which is known as process control block.

Q4. What is Multitasking / Time sharing?

Ans. Multitasking, often referred to as time sharing in computing, is a technique that allows a computer to execute multiple tasks or processes simultaneously by dividing processor time into small slices and rapidly switching between them. This provides the illusion of simultaneous task execution and is a fundamental feature of modern operating systems, improving resource utilization and user productivity.

Q5. Difference betⁿ. user view and kernel view.

Ans. The user view of an operating system is the perspective of the user, who is interested in how the operating system provides services to them. This includes things like the ability to run programs, access files and communicate with other devices.

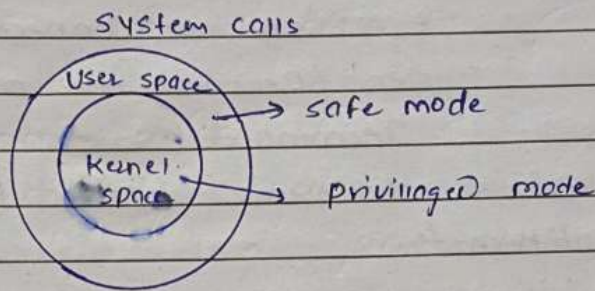
→ The kernel view of an OS is the perspective of the kernel, which is the core of the operating system that is responsible for managing the computer's resources.

User view	Kernel view
A user double-clicks on a file icon to open it.	The kernel allocates memory to the application that is opening the file.
A user presses a key on the keyboard to type a character.	The kernel detects the key press and translates it into a character.
A user clicks on a button in a window to perform an action.	The kernel handles the mouse click and performs the required action.

• Sec-B ($7 \times 3 = 21$)

Q. What is system call? explain various system calls with examples.

Ans. System calls provides an interface to services make available by the operating system.



A system call is a mechanism that provides the interface between a process and the operating system. It is a programmatic method in which a computer program requests a service from the kernel of the operating system.

• Types of System call :- There are five types of system call, that are mention below -

(i) process control :-

creation / Termination of a process

and an abort.

load and execute

get/set process attributes.

(ii) File Management :- open and close, read, write, create and delete a file.

(iii) Device Management :-

Request and release resources

→ these system calls are responsible for device manipulation such as, reading from device buffers, writing into device buffers, etc.

(iv) Information Maintenance :-

get/set Date and time.

get/set system data

get/set process information.

(v) Communication :-

Establish a connection

Send/Receive the message

Terminate.

Q2. What is difference betⁿ Kernel, Process and multithreads.

Ans. A kernel is the core of an operating system. It is responsible for managing the system's resources, such as the CPU, memory and storage. The kernel also provides a set of system calls that applications can use to interact with the system.

→ A process is an instance of a running program. It is a self-contained unit of execution that has its own memory space, code and resources. Processes can communicate with each other and with the kernel through system calls.

→ Multithreading is a technique that allows a process to run multiple tasks simultaneously. Each task is executed by a thread, which is a lightweight process. Threads share the process's memory space and resources, but they have their own stack and program counter.

• Differences :-

Characteristic	Kernel	Process	Multithreading
Definition	The core of an operating system.	An instance of a running program	A technique that allows a process to run multiple tasks simultaneously.
Management	Managed by the operating system.	Managed by the OS.	Managed by the OS, or by a thread library.
Resources	Manages the system's resources, such as the CPU, memory and storage.	Has its own memory space, code and resources.	Shares the process's memory space and resources.
Examples	Device drivers, File system drivers, network drivers.	web browser, word processor, video game.	User interface, background task, database transactions.

Q3. List the services provided by operating system in detail.

Ans. Operating system provide a wide range of services both to user and to application. Some of the most important services include :-

- Process Management :- The operating system is responsible for creating, scheduling and terminating processes. It also provides mechanism for processes to communicate and synchronize with each other.
- Memory management :- The OS manages the system's memory, allocating it to processes as needed. It also provides mechanisms for processes to share memory with each other.

- **File Management :-** The OS provides a file system for storing and organizing files. It also provides mechanism for applications to access and manipulate files.
- **I/O Management :-** The OS provides a layer of abstraction betⁿ. applications and I/O devices, such as disks, printers, and network adapters. This allows application to interact with I/O devices without having to know the specific details of the underlying hardware.
- **Security and Protection :-** The OS provides mechanism for protecting the system's resources from unauthorized access.
- **Networking :-** The OS provides support for networking allowing applications to communicate with each other over a network.
- **User interface :-** The OS provides a user interface for interacting with the system. This interface can be graphical, text-based or a combination of both.
- **Error handling :-** The OS handles error that occurs in the system or in applications.

• Section - C (11x3=33)

1. Explain various types of OS in detail with examples.

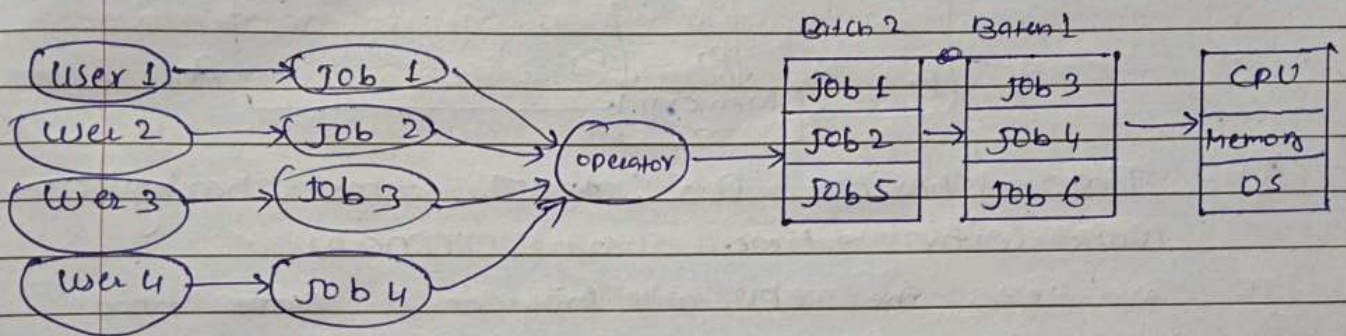
Ans.

Operating system is an interface between the hardware and user. An operating system is responsible for management and coordination of activities, sharing of resources of the computer.

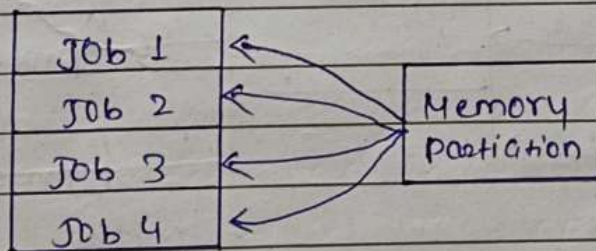
Dec 22, 2023, 17:51 Windows, Linux, Ubuntu, etc.

There are many different types of Operating System, that is following -

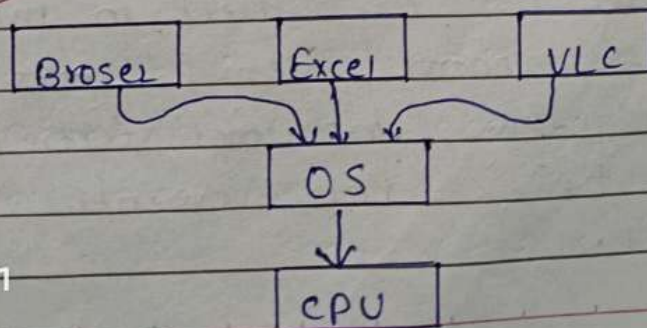
- (i) Batch OS :- The batch operating system will work to submit similar kinds of job together. In this OS, we do not interact directly with our computer system.



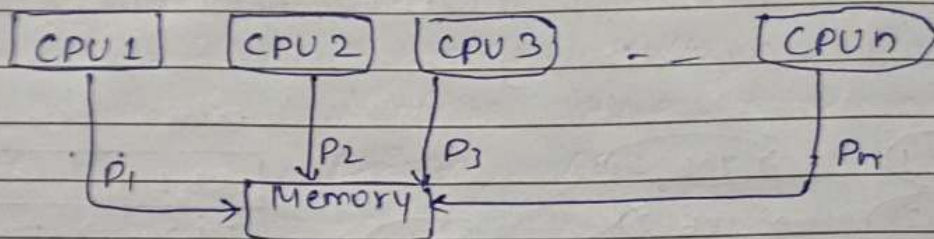
- (2) Multiprogramming OS :- In multiprogramming OS, for the time a process does I/O the CPU can start the execution of other process. Multiprogramming is responsible for parallel processing.



- (3) Multitasking OS :- Multi-tasking OS is simply a multiprogramming OS with having facility of a Round-Robin Scheduling Algorithm. It can run multiple programs simultaneously.

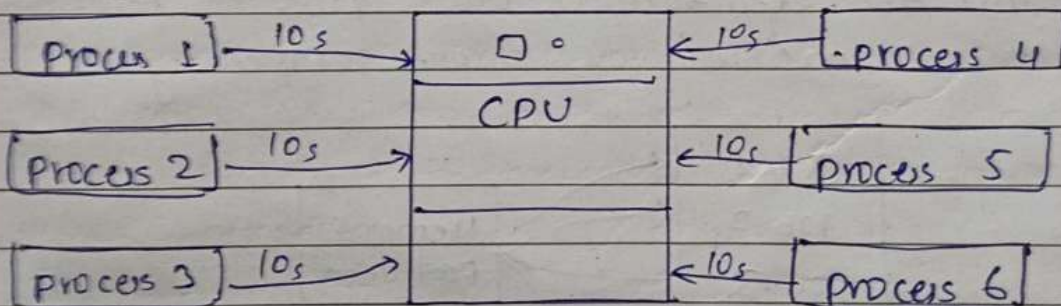


- (4) Multiprocessing OS :- multiprocessing OS is a type of OS in which more than one CPU is used for the execution of resources. It better the throughput of the system.



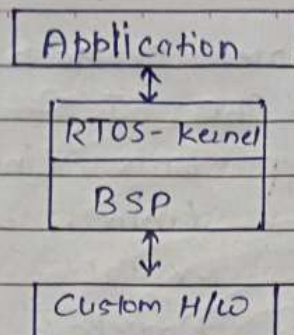
- (5) Time sharing OS :- The time sharing OS work with time sharing concept.

Here, the CPU will provide a same time period to each on every process to complete its tasks as soon as possible whether it a long process or short process.



- (6) Real Time OS :- These types of OS serve real time system. It is very useful when we required a quick response.

→ Real time OS are used in missile systems, air traffic control system, robots, medical imaging system, industrial control system, where time requirements are very strict.

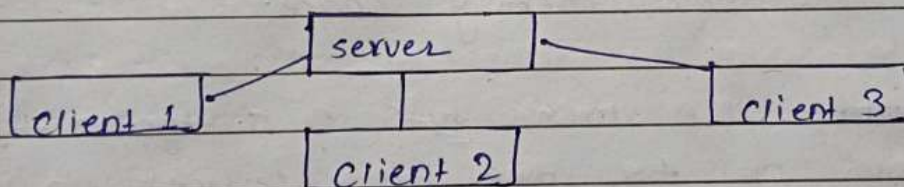


- (7) Embedded OS :- These are the specially design system for specific function in which little or no user interaction is required.

ex'

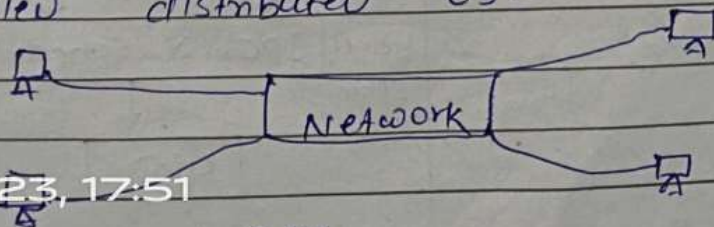
Home Application
ATM Machine
CCTV Camera

- (8) Network / client-server Operating System :-



These system Run on server and provide capability to manage data, groups, users, Application and security. These OS allows shared access to files printer application etc.

- (9) Distributed Operating System :- When many computers are interconnected to each-other through a network for the purpose of sharing their task then it is called distributed OS.



Here some examples of each types of OS :-

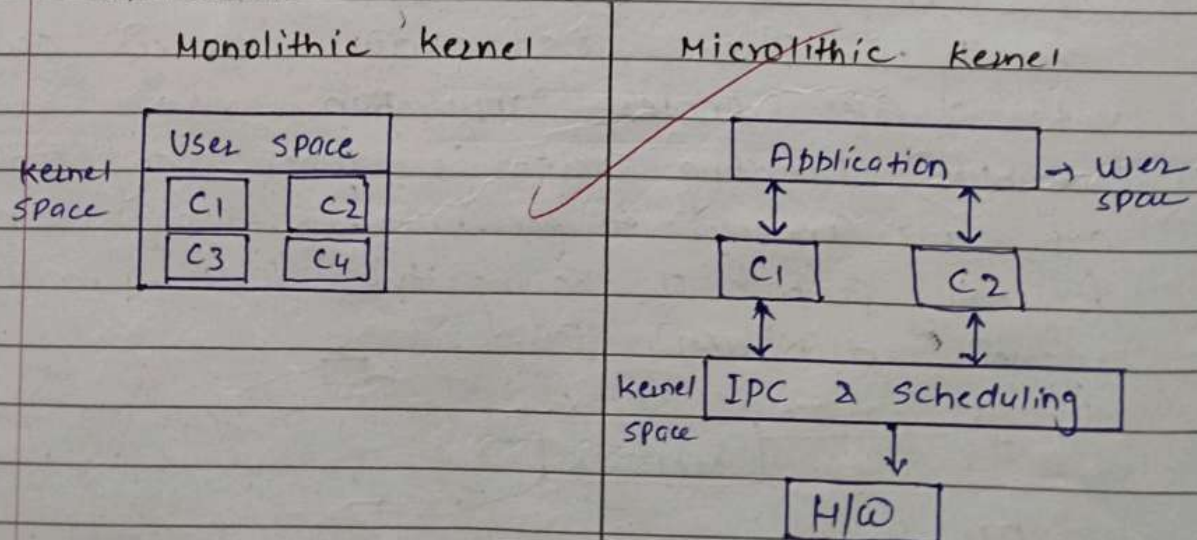
- Batch OS :- OS/360, JES2
- Multiprogramming OS :- Unix, Linux, Windows
- Multiprocessing OS :- Linux, Windows Server
- Real-time OS :- VxWorks, QNX
- Distributed OS :- Amoeba, Mach
- Network OS :- Windows server, Novell NetWare
- Mobile OS :- Android, iOS

Q2. Differentiate between monolithic and microkernel with Diagram.

Ans. Monolithic kernel :- A monolithic kernel is a type of kernel in which all the operating system services run in kernel space, meaning they all share the same memory space.

→ Microkernel :- A microkernel is a type of kernel in which only the most basic services run in kernel space, with other services running in user space.

• Differentiate —





Monolithic kernel

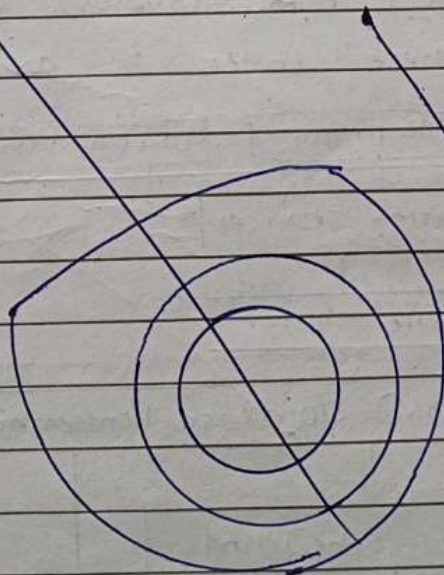
- Kernel size is large.
- It is faster.
- OS is complex to design.
- Difficult to add new functionalities.
- Failure in one component affects entire system.
- Direct function calls within kernel.
- examples:- Windows, Linux, BSD.

Microkernel

- Kernel size is small.
- It is slower compare to monolithic.
- OS is easy to design, implement and install.
- Easy to add new functionalities.
- Failure in one component will not affect entire system.
- Message passing between user-space servers.
- Examples:- L4Linux, macOS.

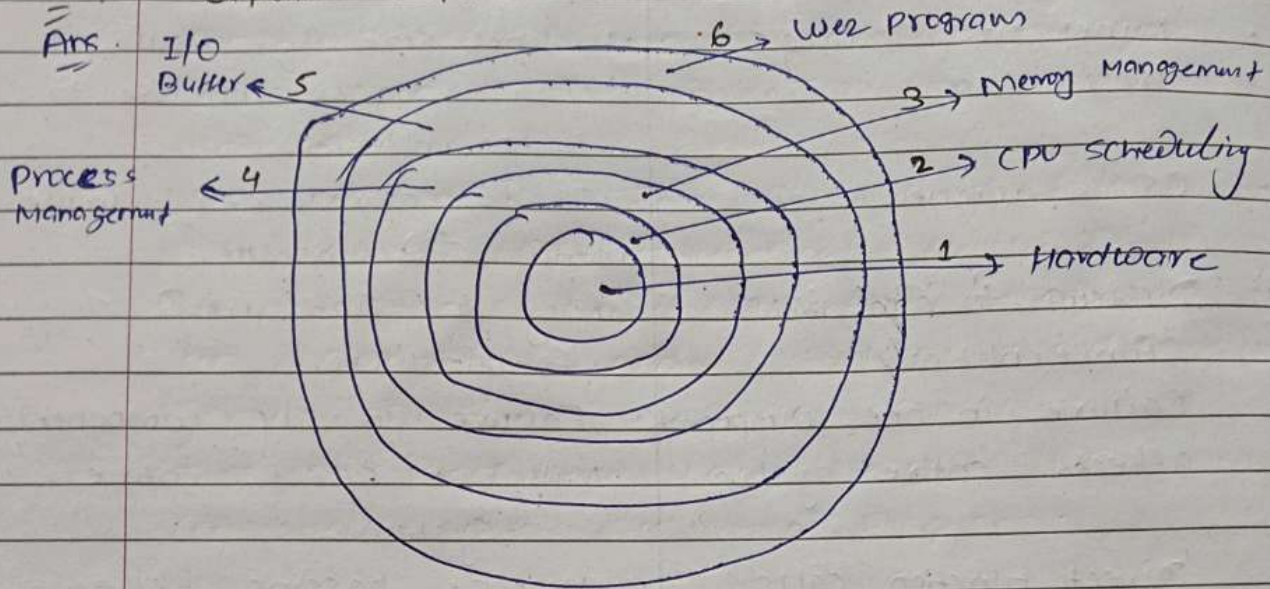
Q3. Explain layered architecture of OS.

Ans.



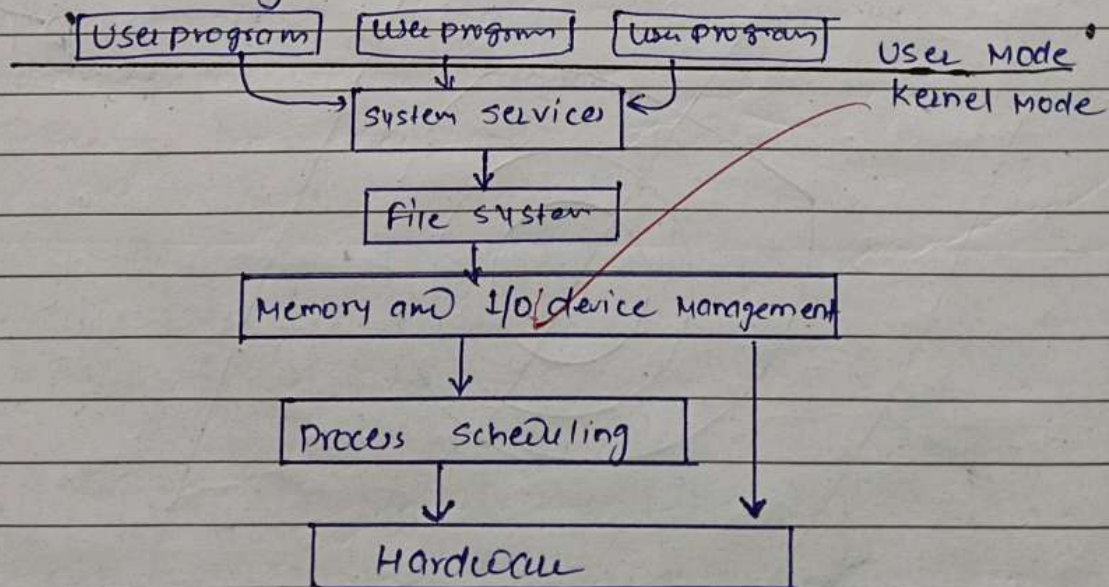
PTD →

Q3 Explain Layered architecture of OS.



The OS can be implemented with the help of various structures. The structure of the OS depends mainly on how the various common components of the operating system are interconnected and merged into the kernel.

→ There are six layers in the layered OS. A diagram demonstrating these layers is as follows:



- (i) **Hardware :-** This layer interacts with the system hardware and coordinates with all the peripheral devices used, such as a Printer, mouse, keyboard, scanner, etc.
- (ii) **CPU scheduling :-** This layer deals with scheduling the process for the CPU. Many scheduling queues are used to handle processes. When the processes enter the system, they are put into the job queue. The processes that are ready to execute in the main memory are kept in the ready queue. This layer is responsible for managing how many processes will be allocated to the CPU and how many will stay out of the CPU.
- (iii) **Memory Management :-** Memory management deal with memory and moving processes from disk to primary memory for execution and back again. This is handled by the third layer of OS. All memory management is associated with this layer.
- (iv) **process Management :-** This layer is responsible for managing the processes i.e., assigning the processor to a process and deciding how many processes will stay in the waiting schedule. The priority of the processes is also managed in this layer. The different algorithms used for process scheduling are FCFS, SJF, priority scheduling, round-robin scheduling, etc.
- (v) **I/O Buffer :-** I/O devices are very important in computer system. They provide user with the means of interacting with the system. This layer handles the buffer for the I/O devices.

and makes sure that work correctly. The computer uses buffers to maintain the good timing speed of the processor and input/output devices.

(vi) User Program :- This is the highest layer in the layered OS. This ^{layer} deals with the many user programs and applications that run in an operating system, such as word processors, games, browsers, etc. You can also call this an application layer because it is concerned with application program.

• Advantages of layered structure :-

- (i) Modularity
- (ii) Easy debugging
- (iii) Easy update
- (iv) No direct access to hardware
- (v) Abstraction

• Disadvantages of layer structure :-

- (i) Complex implementation
- (ii) Slower in execution
- (iii) ~~(iii)~~

20/11/19

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